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NPTEL ONLINE CERTIFICATION COURSE

SUSTAINABLE TRANSPORTATION SYSTEMS

Lecture 17: EIA Case Study-I

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- Case study of Mumbai-Ahmedabad High Speed Rail (MAHSR) project
  - Project details
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  - Monitoring



Mumbai- Ahmedabad High Speed Railway (MAHSR) Project

- First proposed in the Indian Railways' 'VISION 2020' released by the Ministry of Railways, Govt. of India.
- High speed corridors allows speeds up to 350 kmph.
- To use Shinkansen trains from Japan
- Connects Mumbai to Ahmedabad through high speed rail line
- Completion expected by year 2028.



Shinkansen Train

- Shinkansen is popularly known as ‘Bullet Train’.
- Famous for zero causality caused by train accident (in 50 years time with 10 billion passengers)
- Literal meaning of ‘Shinkansen’ is “New trunk/main line”.



Image source: popularmechanics.com

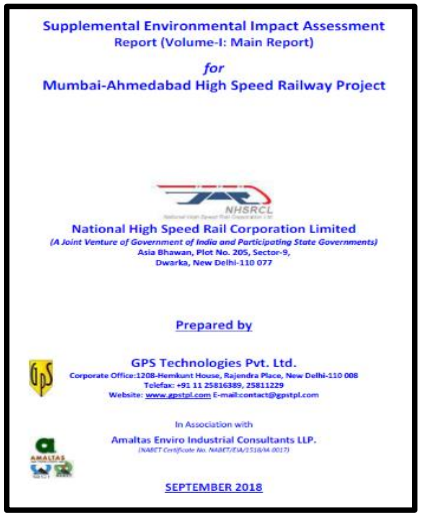
EIA report of Mumbai- Ahmedabad High Speed Railway (MAHSR) Project

Project Proponent: **National High Speed Rail Corporation Limited (NHSRCL)**

Report of EIA prepared by-

- GPS Technologies Pvt. Ltd: S-EIA Report, collection and assessment of primary / secondary environmental data
- NHSRCL / RITES: Land Acquisition Plan and related details
- ARCADIS: Resettlement and Rehabilitation Action Plan

RITES: Rail India Technical and Economic Services
NHSRCL: National High-Speed Rail Corporation Limited


Supplemental Environmental Impact Assessment
Report (Volume-I: Main Report)
for
Mumbai-Ahmedabad High Speed Railway Project

National High Speed Rail Corporation Limited
(A Joint Venture of Government of India and Participating State Governments)
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In Association with
Amaltas Enviro Industrial Consultants LLP.

SEPTEMBER 2018

National High-Speed Rail Corporation Limited (NHSRCL)

- NHSRCL was incorporated on 12th February 2016 under the Companies Act, 2013.
- Finance, construct, maintain and manage the high-speed Rail Corridor in India. modelled as 'Special Purpose Vehicle' in the joint sector
- Equity participation by Central Government through Ministry of Railways and two State Governments viz. Government of Gujarat and Government of Maharashtra.



Special Purpose Vehicle (SPV)

- SPVs are joint ventures and legal entity to fulfill a certain objective/project
- SPVs are generally collaboration of multiple parties to achieve a common goal
- Reduces risk for any single entity
- In India, government has created SPVs for various large scale infrastructure projects.



Project Location & Stations

12 Stations (shown on image)

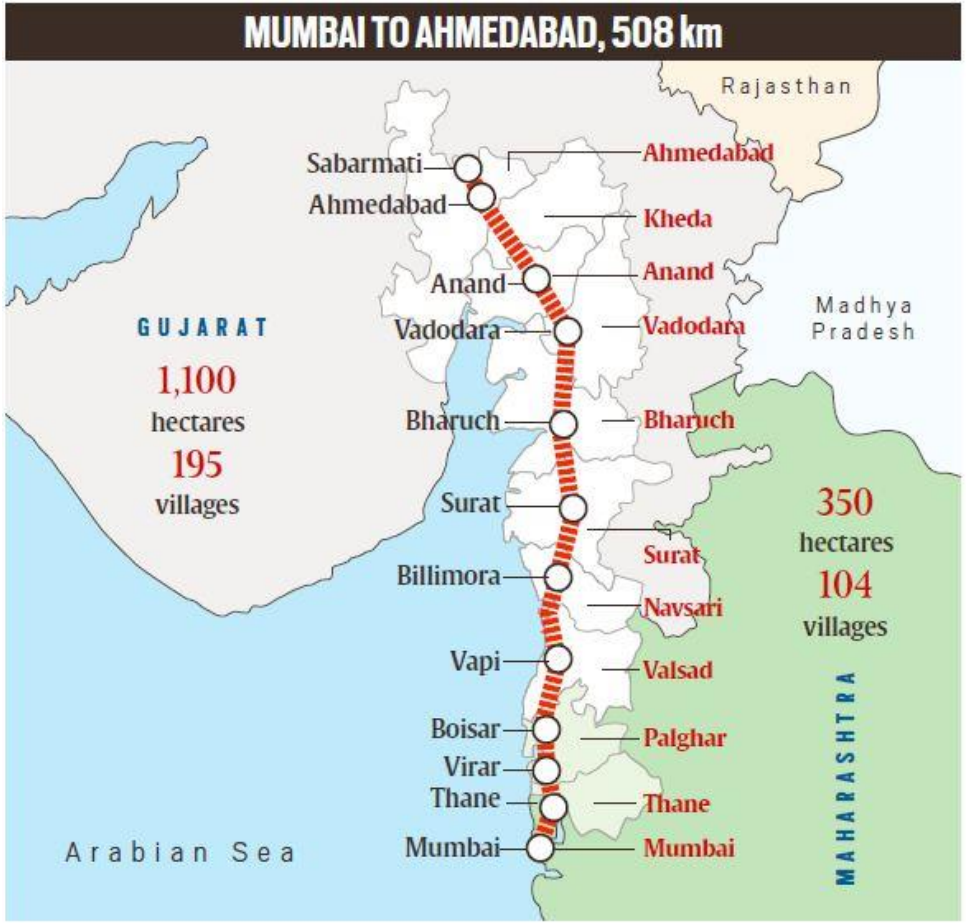


Image: iasparliament.com



Image: Start point of project (BKC-Mumbai)

Major benefits of the MAHSR project

- Reduction in travel time from 6.5 hrs. to 2.5 hrs.
- Major connectivity boost to less developed areas such as Palghar & Valsad and industrial towns like Vapi and Bharuch.
- Technological enhancement of Railways, long term infrastructure development.
- Construction phase will employ about 20,000 people, operations and maintenance 4,000 people & 16,000 indirect jobs.
- Decongestion of conventional rail, road, and air traffic between Mumbai and Ahmedabad
- Reduction in Greenhouse gas emissions due to more efficient transport option.
- Development of State-of-the-Art manufacturing and software ecosystem.



Benefits over other modes of travel

Aspect	MAHSR	Railway/Road/Air
Land Use	2.8 ha/km	9.3 ha/km
	Minimal change in land use pattern	Change at large scale (New roads will be required in future)
Travel Time	2 hr.	6-12 hr.(Rail), 1.25 hr (Air)
	Better connected to local transport	Airports are not easily accessible in many cities
GHG (Greenhouse Gas) emission	10% annual reduction wrt no change	6 times by 20250
	81400 Tons CO2 over 15 years	1.4 Metric Tons CO2/km by 2050
Fuel Consumption	Significant reduction due to mode change	Steep increase if no change

MAHSR: Mumbai-Ahmedabad High Speed Rail

Possible benefits and after comparison w.r.t. other modes, this project can be said 'feasible to implement' (from development point of view)

Financing of project

- Estimated cost of the MAHSR is INR 1.08 Lakh Crores.
- Government of Japan has agreed to provide a soft loan of 12 Billion USD at 0.1% interest.
- Repaid over 50 years and includes a 15-year grace period.
- The loan will be made available through Japan International Cooperation Agency (JICA).

Grace period

- Term generally used in international landings.
- Time period in which, annual payments are not required

Alternative consideration in project planning

3 alternatives were considered:

- Alternative 1- HSR with Viaduct / Bridges and Tunnels
- Alternative 2- HSR with Embankment and Bridges and Tunnels
- Alternative 3 – No-HSR - Business-as-Usual (BAU)

BAU
- Business as usual condition is ‘no change’ situation.

Viaduct Vs. Embankment Rail



Elevated rail section (viaduct)



Rail line on embankment

Image source: railwaymagzine.co.uk, researchgate.net

Comparison among Alternatives

Aspect	Alternative-1 (with Viaduct)	Alternative-2 (with embankment)	BAU (Business as usual)
Length	508.17 km	508.17 km	538 km(Road)
Travel Time	2 hr. 07 min.	2 hr. 07 min.	8 hr. 52 min.(Car)
Travel Demand(by year2023 per day)	40 thousand	40 thousand	38 thousand
Safety Level	High	High	Low
Cost	1080000 Million INR	709151 Million INR	N/A
Environment Impact	Minor damage but Thane creek not affected	Minor damage but Thane creek not affected	N/A
Land Use	2.8 hectare/km	3.2 hectare/km	7.5 hectare/km for conventional rail line

Regulatory Requirements for lender (JICA) for impact assessment

- The MAHSR project is categorized is category A as per JICA's guidelines for environmental and social consideration(April 2010).
- These guidelines are like world bank guidelines for EIA.

- JICA (Japan International Cooperation Agency) is an international lender organization supported by Japan.
- Generally, provides loans on easy terms to countries having good terms with Japan.



Regulatory clearance requirements for project implementation

Clearances related to:

- Diversion of forest land
- Mangroves and wildlife (protected) areas
- Tree felling in various land types
- Coastal regulation zone
- Adverse environment impacts
- Occupational Health & Safety as well as Labor laws



Regulatory authorities for environment in consideration

- Ministry of Environment, Forest & Climate Change (MoEF&CC)
- Central/State Pollution Control Board (CPCB & SPCB)
- National/State Board for Wildlife (NBWL & SBWL)
- Forest Department (State- Gujarat & Maharashtra)
- National Green Tribunal (NGT)
- National/State Coastal Zone Management Authority (NCZMA & SCZMA)
- Dahanu Taluka Environment Protection Authority (DTEPA)

DTEPA

Taluka level authority in Dahanu; created to protect natural habitat of tribal people in tribal district of Palghar.

Environmental Acts considered for assessment

- Environment (Protection) Act 1986
- Air (Prevention and Control of Pollution) Act and Rules 1981
- Water (Prevention and Control of Pollution) Act and Rules 1974
- The Indian Forest Act 1927
- Wildlife (Protection) Act 1972
- Maharashtra Felling of Trees (Regulation) Act 1964
- Maharashtra (Urban Areas) Protection and Preservation of Trees Act 1975
- The Wetlands (Conservation and Management) Rules 2010

Environment attributes considered in MAHSR project

- Physiography & terrain
- Drainage
- Soil
- Geology & seismology
- Landscape
- Land use
- Ground subsidence
- Climate
- Climate change
- Water use/demand
- Water quality
- Water body
- Air quality
- Noise
- Vibration
- Ecology
- Protected areas
- Sensitive location
- CRZ (Coastal Regulation Zone)
- Forest & mangroves

Landscape: Impact & mitigation measures

Impact	Mitigation
• Construction of viaduct from Mumbai to Ahmedabad • Bridges across the intersecting water bodies • Permanent change in the vicinity of the proposed stations	• Suitable (less impact to landscape) viaduct design was chosen • Landscaping is effectively managed and minimize temporary visual impacts during construction • Reduce the impact of the project and structures by additional tree planting wherever feasible to reflect the existing landscape character. • Landscaping at the station to improve aesthetics etc.

Deforestation: Impact & mitigation

Impact	Mitigation
• Diversion of forest (138.3269ha. including 32.3902 ha. Of Mangrove). • Removal of mangrove trees 24.3981ha.) • 60000 trees are expected to be cut	• Compensatory afforestation and planting of mangroves and trees shall be carried out as per regulatory requirement. • Construction of balancing culverts in coastal and mangrove areas, to maintain tidal water exchange.



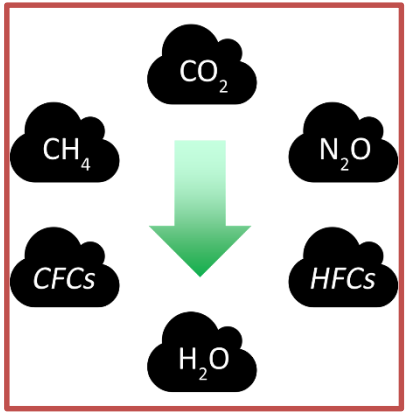
Rise in Sea level: Impact of project & mitigation

Impact	Mitigation
• Impact of potential sea level rise on the alignment and station structures close to coastal area at BKC-Mumbai	• The structure and premises of proposed Station close to Mithi river designed taking into consideration of sea level rise of 1 m over a period of 100 years.

Annual global sea level rise is 3.6 mm per year in last decade

Energy consumption & GHG emissions: Mitigation measures

Impact	Mitigation
• Consumption of electrical energy for operation and at stations • Associated indirect GHG emissions	• The Shinkansen HSR technology is relatively more efficient as compared to the conventional railway operation presently in use in India • Energy star rated equipment and fixtures will be used to minimize the energy consumption (energy efficient motors and pumps, use of energy efficient lighting, energy efficient luminaires, adequate and illumination levels optimized as per tasks, and energy) • offset to a large extent by the savings in GHG emissions due to expected modal shift from private vehicle road travel



Dust & emissions during construction: Impact & mitigation

Impact	Mitigation
• Gaseous emissions from construction vehicles • Fugitive dust emissions due to traffic movement • Spillage and fugitive emissions arising out of construction materials	• All the vehicles shall have pollution Under Control (PUC) certificate • Regular sprinkling of water on unpaved ways / access roads • The material shall be transported in covered • An enclosure or barricading at such locations

Noise during construction: Impact & mitigation measures

Impact	Mitigation
• Noise emissions from construction equipment. • Noise generated from the running of heavy vehicles deployed in the construction activities.	• Adequate enclosures around the stationary equipment. • Noise barriers of suitable specifications erected in locations of sensitive receptors. • Equipment fitted with exhaust mufflers • Personal protective equipment (PPE) shall be provided to the workers working near the high noise area.



Vibration during construction: Impact & mitigation measures

Impact	Mitigation
• Increase in ground vibration due to underground tunneling and potential impact on TCFS • Noise propagation due to running of heavy construction vehicles at the construction sites	• Noise abatement measures shall be adopted in tunneling activity. • Noise barriers of suitable specifications shall be erected in locations of sensitive receptors.

Impact on safety & comfort of local community during construction & its mitigation

Impact	Mitigation
• Vibration in the nearby area due to piling.	• Night-time pilling activity avoided near residential areas.
• Blasting operation may lead to temporary ground borne vibration.	• Blasting carried out in the day-time only, with advance intimation to local communities.
• Running of heavy construction equipment.	



Mitigation measures for Flora

- Verification with forest department officials and marking of trees which need to be cut (it minimizes the tree cutting during construction)
- Appropriate compensatory plantation using suitable native species or pollution tolerant species
- For trees to be felled in private land, compensation for trees based on fruit yield value, timber value given to the owners
- Minimizing clearing to only necessary amount of vegetation from the forest area.
- Compensatory mangrove plantation



Image: Flamingo near Thane creek
Source: timesofindia.com

Mitigation measures for Fauna

- No construction yard should be set up in the forest areas
- Heavy construction equipment/machineries should be kept away from the forest areas
- Soil compaction for embankment work should be done immediately to avoid erosion
- Immediate actions for speedy cleaning up of oil spills, fuel and toxic chemicals in the event of accidents
- Crossing structures (tunnels for animal movement) in place where the MAHSR line passes through the forest patches



Image: Mangrove area near Thane creek
Source: hindustantimes.com

Mitigation measures for water bodies/natural drainage

- No material to be dumped into natural drains or block, impede or alter drainage channels.
- Adequate cross drainage channels (longitudinal and median drains) should be provided at suitable locations for the smooth passage of the surface runoff to prevent flooding.
- Capacity of existing drainage works, cross drainage structures in the parallel section should be duly augmented.
- Suitable drainage at construction site and camp to avoid formation of stagnant pool of water that lead to soil erosion, water logging.

Public Consultation

Public meeting were arranged at 10 places:

- Navsari
- Surat
- Valsad/Dadra Nagar Haveli
- Palghar
- Mumbai/Thane
- Vadodara
- Anand
- Kheda
- Ahmedabad
- Baruch


Ahmedabad


Kheda

Compensation and religious structures were key issues discussed by public

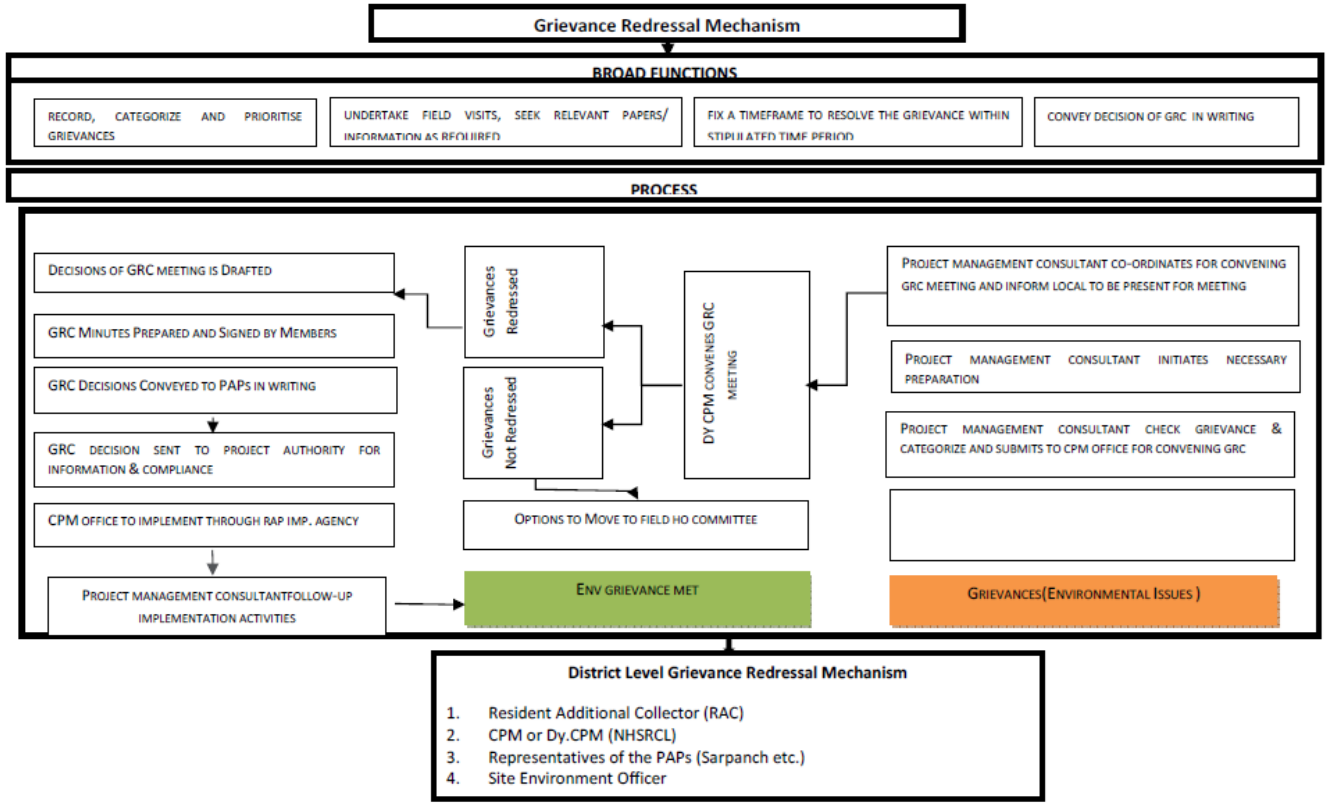
Grievance redressal mechanism

As project passes through many protected areas or tribal region, it makes project exposed to social acceptability issues such as interaction with:

- NGOs/activists for environment
- Organizations for tribal rights
- Local people whose land has been acquired

This makes grievance redressal very import for the project.

Grievance redressal mechanism



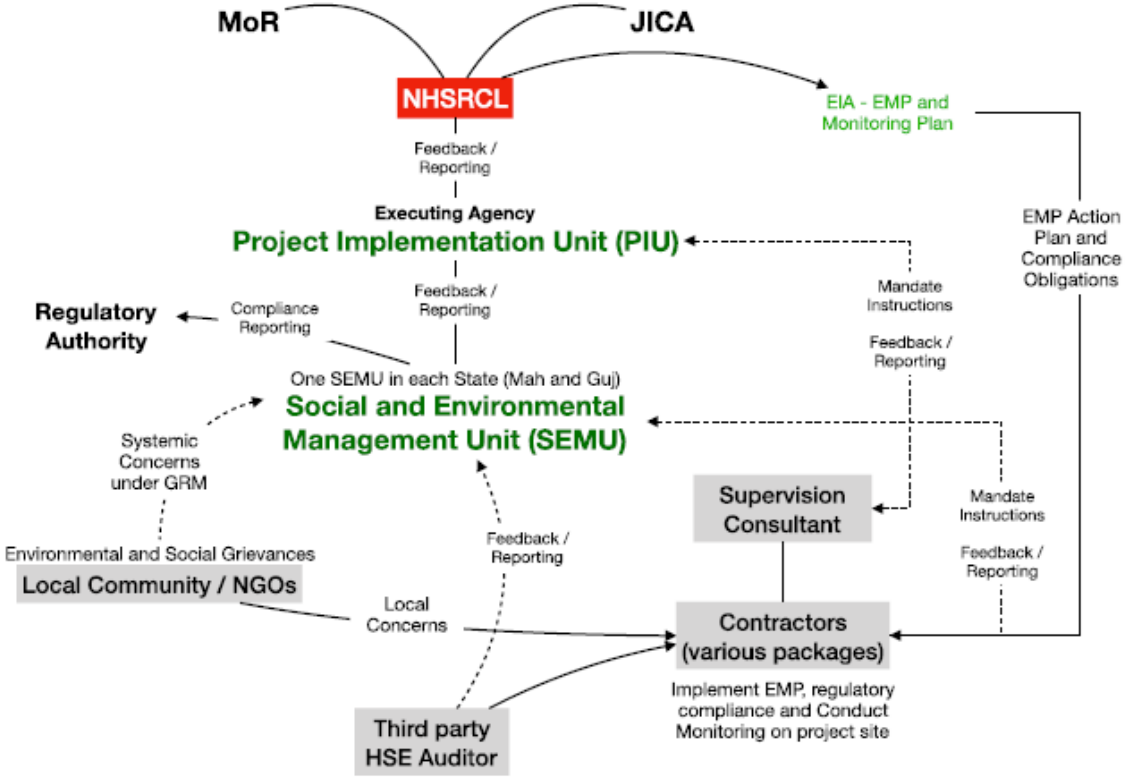
Grievance redressal is also part of public participation process.

Impacts considered for monitoring during operation phase

- Social impacts
- Impacts on crop & vegetation
- Air quality
- Noise & vibration
- Water quality
- Impacts on flora and fauna



Implementation of EMoP (Environment Monitoring Plan)



Mitigation plans and monitoring will be executed by SEMU (Social & Environmental Management Unit).

Inferences of EIA of MAHSR project

- Timeline for completion of project is shifted to 2028 due to delays in land acquirement.
- Project passes through eco-sensitive coastal zone and tribal areas.
- Mitigation measures and strict monitoring of impacts should be done to minimize adverse impacts.



Conclusion

- Binding & regulatory EIA process ensures
  - Impact assessment
  - Mitigation for impacts considered
  - Use of practices which are less adverse for environment
  - Monitoring of impacts and mitigations measures
  - Public participation during project

- In India, EIA process follows various steps which help in minimizing the adverse environmental impacts from developmental activities.

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Thank You